

Do Financial LLM Agents Discover New Alpha?

A 98-Work Audit and 50-Strategy Factor-Spanning Study

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Abstract

Financial LLM agents often produce strategies that look profitable, but apparent profitability is not the same as alpha incremental to a broad factor benchmark. We screen 98 distinct works, retain 69 formula-discovery or trading papers, and construct 50 good-faith U.S. strategy reconstructions for all 40 retained works with enough public content for a component or motif mapping; the other 29 remain availability-only. Holding returns, costs, and evaluation months fixed, median annualized one-month-ahead residual return falls from 9.97% under CAPM to 3.17% under FF3, 2.69% under FF5 plus momentum, and -1.69% under the market plus 132 Jensen-Kelly-Pedersen (JKP) factors. The corresponding numbers of Holm-adjusted positive results are 6, 1, 2, and 0 out of 50. The median reconstruction's strongest absolute correlation with a JKP factor is 0.81, and 47 of 50 exceed 0.50. Among 38 mapped papers with verified full text, none reports a factor-adjusted intercept and none uses JKP132. These results support a bounded interpretation: within these researcher-implemented common-task reconstructions, the broad factor library accounts for much of the return variation that appears as alpha under narrower models, and the public evidence does not establish incremental return information for this reconstructed family. The analysis does not show that all screened papers lack methodological novelty or practical value, nor that unavailable native agents would produce the same returns. Because the reconstructions and benchmark ladder were designed after U.S. outcomes were inspected, the estimates are descriptive spanning diagnostics, not confirmatory tests of native-agent performance.

CCS Concepts

• **Computing methodologies** → **Multi-agent systems**; • **Applied computing** → **Economics**; • **General and reference** → *Empirical studies*.

Keywords

financial agents, alpha, asset pricing, factor zoo, reproducibility, multiple testing

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1 Introduction

LLM-agent papers increasingly report profitable factors, superior trading performance, or investment alpha [14, 41, 64, 69, 93]. Many of those strategies may indeed earn positive returns relative to

the papers' chosen baselines. The central empirical question, however, is whether they add return information beyond familiar asset-pricing factors. Positive returns can coexist with exposures to low-risk, momentum, quality, or valuation factors, so separating incremental return information from familiar exposure requires an explicit spanning analysis.

Our evidence separates these claims. Across 50 reconstructions from 40 retained papers, apparent alpha remains positive against CAPM, FF3, and FF5 plus momentum, but collapses when the benchmark expands to the market plus 132 JKP characteristic factors [23]. Median annualized residual return moves from 9.97% under CAPM to -1.69% under the broad model, and the Holm-adjusted discovery count falls from six to zero. The return series themselves point to the same explanation: 47 of 50 have an absolute correlation of at least 0.50 with their nearest JKP factor, and the median nearest-factor correlation is 0.81.

This pattern is consequential because the source papers generally do not ask the spanning question. Of the 38 mapped papers whose full text we verified, 37 report no asset-pricing regression, one reports multifactor loadings without an intercept, 0 report factor-adjusted alpha, and 0 use JKP132. Consequently, return, Sharpe, predictive, or narrow-factor comparisons do not by themselves establish that a strategy is incremental to a broad anomaly library. The paper does not judge the scientific quality, methodological novelty, or practical value of all 98 screened works, and it does not infer failure for unavailable native agents. It asks the narrower question of whether the 50 disclosed common-task reconstructions retain alpha relative to JKP132.

One possible explanation for the observed overlap is that many systems prompt, retrieve, or search over financial concepts and symbolic factor expressions [41, 61, 93]. Established anomaly ideas are common in financial text and factor libraries. Our design cannot determine whether any agent learned, retrieved, independently rediscovered, or arrived by another route at such content: the benchmark ladder and correlation map describe the reconstructed return paths, not the causal origin of an agent's strategy.

Contributions. We make four contributions. First, we provide a cutoff-bounded 98-work evidence waterfall that distinguishes 69 retained works, 40 reconstructed works, and 29 availability-only works without imputing failures as zero returns. Second, we evaluate all 50 retained reconstructions on one matched benchmark ladder, from CAPM through JKP132, rather than numerically back-testing only the five papers with the strongest source grounding. Third, we map every reconstructed strategy return series to all 132 JKP factors and summarize the factor families most closely aligned with those return paths. Fourth, we audit the benchmarks used in the mapped source papers and separate the primary factor-spanning result from a secondary 14-attempt reproducibility audit and other sensitivity analyses.

One-sentence takeaway. For the 50 post-hoc reconstructions, alpha under narrow benchmarks largely attenuates under JKP132 and no familywise-positive residual remains; this concerns the reconstructions, not the novelty or value of the source papers or unavailable agents.

2 Evidence Design

2.1 Corpus, denominators, and evidence routes

The screen covers database inception through 2 August 2026. Three query families combine LLM/agent terms with alpha or strategy discovery; trading, portfolios, returns, or backtests; and benchmark, audit, artifact, or leakage terms. Sources are arXiv’s relevant CS and q-fin categories, ACL Anthology, OpenReview, ACM DL, and SSRN, plus citation chasing from AlphaBench, AlphaQT-Bench, FINSABER, and the Agentic Trading survey. Crossref/OpenAlex resolve duplicates; GitHub locates only record-linked artifacts. Raw hit counts and rankings were not preserved, so this is a cutoff-bounded systematic screen rather than a complete universe.

The registry has 103 lineages, 104 record URLs, and 98 works after six duplicate manifestations are collapsed. All 98 works are cited. Frozen rules retain 30 formula-discovery and 39 trading works [4, 9, 14, 25, 28, 41, 46, 49, 57, 61, 64, 67, 69, 72, 83, 91] [3, 20, 29, 39, 42, 45, 60, 65, 71, 73, 85, 87, 92, 93] [15, 40, 68, 76, 89, 90, 95] [1, 2, 7, 17, 18, 27, 36–38, 43, 44, 47, 51, 54, 62, 63, 66, 74, 75, 78, 79, 97] [5, 6, 11, 16, 22, 24, 33, 35, 56, 70] and exclude 29 benchmarks, audits, community tools, or mechanistic comparators with recorded reasons [10, 58, 59, 80, 88, 94] [8, 12, 13, 31, 32, 52, 82, 86] [19, 26, 30, 34, 48, 50, 53, 55, 77, 81, 84, 96, 98–100]. Thus 98/69/40/50 denote screened works, retained works, reconstructed works, and reconstructed strategies; they are not competing replication counts.

Before mapping, each retained work takes one mutually exclusive route: public code (native execution or a precise blocker precedes any secondary proxy), a sufficiently specified paper-only procedure (reproduce only the stated prompt/search/training scope), or an underspecified paper-only procedure (use a labeled component/motif proxy, otherwise availability-only). The route ledger assigns 18, 0, and 51 works, respectively. Thus no paper-only work is claimed as a full procedure replication: among the latter, 3 support partial components, 24 motif proxies, and 24 are availability-only.

We construct 50 mappings for 40 retained works. Five works support 13 source-grounded component tests, while 35 works support 37 favorable in-spirit reconstructions. The other 29 retained works remain availability-only because no frozen common-panel mapping is supportable; we neither invent a proxy nor assign zero alpha. One of the 50 strategies adapts a released QuantEvolver seed expression, 12 reconstruct source-grounded components, and 37 translate higher-level narratives. Every estimate inherits that provenance: a reconstruction tests the historical economic content we implemented, not the unavailable native agent.

2.2 Common U.S. task

The monthly U.S. panel runs from August 2001 through December 2024. At formation month t , the universe $\mathcal{U}_t$ contains the 1,000 largest positive-market-equity common stocks, and mapping i produces point-in-time score $s_{ijt} = g_i(x_{jt})$. For the standard value-weighted long–short mapping, with top and bottom deciles H_{it} and

L_{it} and market equity $m_{jt} > 0$, target weights are

$$w_{ijt} = \begin{cases} m_{jt} / \sum_{k \in H_{it}} m_{kt}, & j \in H_{it}, \\ -m_{jt} / \sum_{k \in L_{it}} m_{kt}, & j \in L_{it}, \\ 0, & \text{otherwise.} \end{cases} \quad (1)$$

Labeled long-only and sparse top- k mappings preserve their stated rules. Formation never conditions on next-month return availability. Missing held returns receive zero in the primary analysis without renormalizing surviving weights; an adverse unit-move stress is reported separately.

Turnover restores target exposure from drifted pre-trade weights. With total security return $R_{j,t}^{\text{tot}}$ and held excess return $R_{j,t+1}^{e,*}$,

$$\begin{aligned} \tilde{w}_{ij,t-} &= \frac{w_{ij,t-1}(1 + R_{j,t}^{\text{tot}})}{1 + \sum_k w_{ik,t-1} R_{k,t}^{\text{tot}}}, \quad \tau_{it} = \sum_j |w_{ijt} - \tilde{w}_{ij,t-}|, \\ r_{i,t+1}^{\text{net}}(c) &= \sum_j w_{ijt} R_{j,t+1}^{e,*} - c \tau_{it}, \quad c \in \{0, 5, 10, 25, 50\} \times 10^{-4}. \end{aligned} \quad (2)$$

The primary one-way rate is 10 bp. This linear cost model omits borrow, financing, nonlinear impact, and capacity; gross benchmark factors make the comparison conservative.

2.3 Matched benchmark ladder and factor overlap

The headline analysis restricts attention to all 50 strategies associated with the 40 reconstructed retained works and fixes their net returns, 126 evaluation months (August 2011–January 2022), and inference across four benchmarks. For benchmark b , slopes use a rolling 120-month window. The broad model contains the market and 132 JKP characteristic returns; the six FF5-plus-momentum analogues remain unpenalized, and ridge penalizes the other 127 slopes. The penalty $\lambda \in \{0.1, 1, 10, 100, 1000\}$ minimizes mean-squared error over the last 24 months of each training window before refitting on all 120 months. CAPM, FF3, and FF5 plus momentum use unpenalized rolling OLS. For each test month,

$$\begin{aligned} \hat{\theta}_{it}^{(b)}(\lambda) &= \arg \min_{\theta=(a,\beta)} \left\{ \sum_{s=t-120}^{t-1} (r_{is} - a - \beta^\top f_s^{(b)})^2 \right. \\ &\quad \left. + \lambda \|P_b \beta\|_2^2 \right\}, \\ e_{it}^{(b)} &= r_{it} - \hat{\beta}_{it}^{(b)\top} f_t^{(b)}, \quad \hat{\alpha}_i^{(b)} = 12 \bar{e}_i^{(b)}, \end{aligned} \quad (3)$$

where P_b exempts the six base returns from penalization. Omitting the fitted intercept from $e_{it}^{(b)}$ makes its mean the factor-unexplained return. HAC statistics use $L = 4$; Holm and a 5,000-replication moving-block maximum- $|t|$ procedure control the 50-strategy family conditional on the mappings.

To identify familiar economic content directly, we correlate each strategy with every JKP factor over the 246 common months:

$$\rho_{ik} = \text{Corr}(r_i, f_k^{\text{JKP}}), \quad k_i^* = \arg \max_{1 \leq k \leq 132} |\rho_{ik}|. \quad (4)$$

This nearest-factor statistic is descriptive. Correlation does not identify the agent’s inputs or prove that a source paper copied a factor; it tests whether the reconstructed return path resembles a known factor return.

Table 1: The five papers underlying the 13 source-grounded component tests. “Source-grounded” applies to the named component, not the full native agent or original evaluation. The other 37 retained strategies are explicitly labeled in-spirit reconstructions.

Source (<i>N</i> mappings)	Original idea and task	Component implemented in our common task
EFS [49] (3)	LLM evolutionary factor search for sparse top- <i>m</i> portfolios on Fama–French and US50/HSI45/CSI300 datasets.	Momentum/low-volatility/breakout score; reversal/low-noise score; and a sparse equal-weight top-five variant.
QuantEvolver [93] (1)	Reinforcement fine-tuning generates executable factor-DSL expressions across market benchmarks.	Released risk-adjusted-momentum seed $\text{ret}_{12,1}/ \text{rvol}_{252} $; portfolio formation remains ours.
Alpha-Jungle [61] (2)	LLM-guided MCTS searches symbolic stock factors using price, volume, and volatility operators.	Price/volume/momentum and trend-after-volatility-compression scores based on documented examples.
FAMA [41] (1)	Neural-symbolic factor mining evaluated by RankIC/RankICIR and an S&P 500 simulation.	Interpretable value–momentum–profitability score motivated by the paper’s examples.
GuruAgents [27] (6)	Prompt-guided agents emulate named investor philosophies in a Nasdaq-100 backtest.	Graham, Buffett, Greenblatt, Piotroski, and Altman screens plus their ensemble.

3 Results

3.1 Broad controls attenuate reconstructed-strategy alpha

Figure 1 states the paper’s main result. Under CAPM, 44 of 50 strategies have positive estimated alpha, 28 are nominally positive, and 6 survive Holm; median annualized alpha is 9.97%. FF3 reduces the median to 3.17% and leaves 1 Holm result. FF5 plus momentum yields median 2.69% and 2 Holm results. Once the benchmark includes JKP132, the median becomes -1.69%: only 17 estimates remain positive, one is nominally positive, and none survives Holm or maximum- $|t|$ adjustment. The median attenuation from FF5 plus momentum to JKP132 is 4.95% per year.

The result is not driven only by the 37 in-spirit translations. Among the 13 released-expression or source-grounded component strategies, 10 have positive FF3 alpha and 11 have positive FF5-plus-momentum alpha, but only 4 remain positive under JKP132. The benchmark expansion therefore changes the measured residual-return distribution even for the highest-fidelity subset. It also clarifies the role of the five papers in Table 1: they define an attribution tier, not the numerical backtest denominator.

3.2 Which known factors align with the reconstructed returns?

The correlation map supplies the missing economic interpretation. Across the 50 strategies, the median $|\rho_{ik_i^*}|$ is 0.81; 47 have a nearest-factor absolute correlation of at least 0.50. Although 21 different JKP characteristics appear as a closest match, Table 2 shows concentration in low-risk, 52-week-high, volatility, quality, momentum, and valuation families. Betting-against-beta alone is the nearest analogue for 12 strategies and 52-week-high proximity for eight.

The signs are economically coherent. Risk-aware and stable-quality reconstructions often align with low-beta or low-volatility returns. Their correlations with positively oriented volatility factors can therefore be negative. Trend and breakout reconstructions align positively with 52-week-high and momentum factors, while fundamental-style reconstructions align with quality or valuation. These patterns show that JKP132 contains return series closely related to the economic motifs implemented in the common-task mappings; they do not identify what any native agent learned or generated.

The source benchmark audit shows that the mapped papers generally do not report a directly comparable spanning analysis.

Among 38 verified mapped papers, 37 contain no asset-pricing factor regression and 1 reports loadings without a factor-adjusted intercept; none reports factor-adjusted alpha or uses JKP132. Source evaluations instead emphasize predictive scores, cumulative or excess returns, Sharpe ratios, buy-and-hold comparisons, agent baselines, or factor loadings. Those outcomes can establish usefulness relative to a chosen task, but they do not by themselves establish incremental alpha beyond the factor zoo.

3.3 Fidelity, costs, and secondary diagnostics

The longer 2001–2024 six-factor exercise provides an implementation sensitivity, not the headline matched ladder. Among the 13 source-grounded component tests, median annualized six-factor alpha is 0.88%; one is nominally positive and none survives Holm within that subset. Across the broader 62-row ledger, which adds 12 mappings from excluded diagnostic sources, median alpha at 10 bp is -0.05% and one researcher-authored AlphaAgent-inspired quality proxy survives conditional Holm. That result is not agent output, disappears under some fixed HAC choices, and has a simultaneous lower bound of only 0.48%.

Trading costs weaken rather than create the result. Median six-factor alpha falls from 2.15% gross to 1.01% at 5 bp and -0.05% at 10 bp; positive counts across 0/5/10/25/50 bp are 46/42/30/18/10. An adverse missing-return stress leaves zero nominal or Holm positives. A separate 133-factor diagnostic over the 62-row ledger yields one nominal result but zero Holm or maximum- $|t|$ results.

The international extension is forensic rather than performance evidence. Thirty-five paths cross nonpositive NAV; security-level reconstruction shows that every event is dominated by one extreme positive return held short. Without independent validation of those returns and corporate actions, we retain the event ledger but exclude international alpha from the paper’s conclusion.

4 Interpretation and Limits

The evidence supports a bounded statement about the implemented mappings. LLM-agent methods may produce economically meaningful returns and may contribute methodological or practical value through factor discovery, formula search, or portfolio construction. In this study, however, the reconstructed family has negative median residual return against JKP132 and no multiplicity-adjusted positive result. The correlation evidence relates that attenuation to familiar factor families rather than treating a larger benchmark as an unexplained statistical black box.

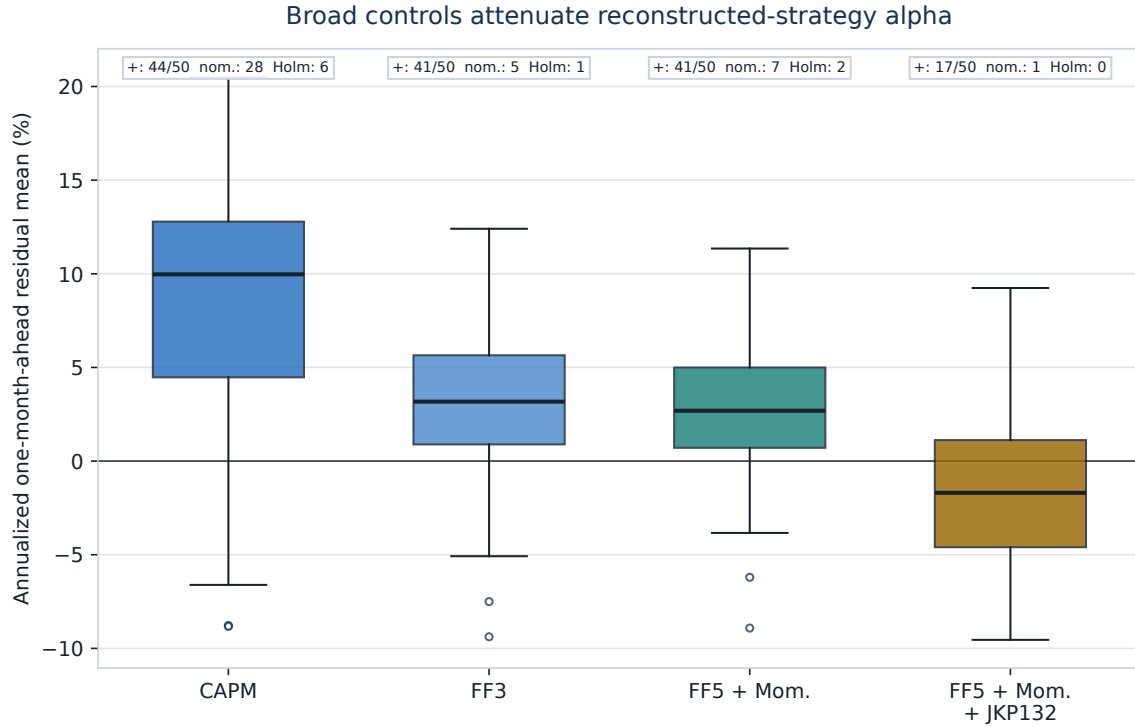


Figure 1: Matched benchmark ladder for all 50 retained strategy reconstructions from 40 papers, net of 10 bp one-way costs. Candidate returns and the 126 evaluation months are identical across models. CAPM, FF3, and FF5 plus momentum use rolling OLS; the broad specification leaves the six base factors unpenalized and ridge-controls the remaining JKP factors. Annotations report positive estimates, positive two-sided HAC tests at 5%, and positive Holm-adjusted tests. Because the mappings and ladder are post-hoc, “one-month-ahead” describes factor fitting, not prospective agent discovery.

Table 2: Most frequent JKP132 analogues of the 50 retained strategy reconstructions. The table reports the JKP factor with the largest absolute return correlation for each strategy, not a factor label claimed verbatim by the source paper.

Closest JKP analogue	Strategies	Median ρ	Median $ \rho $
Betting-against-beta	12	-0.84	0.84
52-week-high proximity	8	0.85	0.85
Realized volatility	3	-0.82	0.82
Idiosyncratic volatility	3	-0.78	0.78
Quality-minus-junk safety	3	0.77	0.77
12–1 momentum	3	0.71	0.71
Downside beta	2	-0.19	0.70
EBITDA/enterprise value	2	0.69	0.69
3–1 momentum	2	0.66	0.66
Average five highest daily returns	1	-0.84	0.84

Correlations use 246 common monthly returns from August 2001 through January 2022. Negative ρ denotes exposure opposite to the published JKP factor orientation. These ten analogues cover 39 strategies; eleven additional JKP factors are the closest match for one strategy each.

Any explanation for the overlap remains a hypothesis. Financial LLM systems operate on pretrained knowledge, prompts, retrieved text, hand-specified operators, or factor DSLs that can foreground established anomalies. When evaluation uses buy-and-hold, Sharpe, predictive metrics, or FF3/FF5, relevant broad-factor exposure may remain unmeasured. Our audit documents the absence of comparable broad adjustment in the verified source papers and measures

overlap in the reconstructed returns. It does not observe proprietary pretraining corpora, retrieval logs, or causal pathways, and it does not attribute any strategy to memorization, retrieval, or rediscovery.

Three limitations prevent a stronger agent-level claim. First, 37 of 50 strategies are favorable in-spirit reconstructions, 12 are source-grounded component tests, and only one adapts released code;

none is a native-agent return replication. Second, mappings and the benchmark ladder were frozen after U.S. outcomes had been inspected and had no independent second coder. Holm and bootstrap procedures cover sampling multiplicity across realized formulas, not mapping discretion, prior factor mining, or publication selection [21]. Third, common-task portfolios omit language, memory, retrieval, debate, native timing, dynamic allocation, non-equities, and execution. A reconstruction failure cannot prove native-agent failure, and reconstruction success cannot be attributed to an agent.

5 Reproducibility and Secondary Code Audit

The secondary repository audit targets 14 implementations: eight retained systems and six excluded diagnostic works. It reproduces zero native agents. One released QuantEvolver seed expression supports a testable code-backed adaptation with annualized six-factor alpha of 0.33% ($t = 0.65$); the other 13 attempts have recorded output, input/adaptor, task, licensing, or executability blockers and do not enter a performance denominator. Keeping this audit secondary avoids conflating repository availability with the 40-paper/50-strategy spanning analysis. This 14-target execution inventory is narrower than the paper-level route audit: among the 18 code-available retained works, 8 have targeted records and 10 have precise static common-task blockers but were not execution targets. In both cases, any common-task proxy is secondary.

The artifact ledger contains the search protocol and 98 dispositions; the 50 strategy definitions and provenance fields; the source-paper benchmark coding; all 6,600 strategy-JKP correlations; the four-model benchmark panel; cost, missing-return, and multiplicity outputs; and the 14 code-attempt records. Deterministic builders verify hashes and regenerate tables and figures. Tests cover timing, weights, drift, costs, factor alignment, rolling estimation, denominators, correlation completeness, and claim values.

Responsible use and AI assistance. No orders were transmitted. Results are retrospective research evidence, not investment advice. Generative AI assisted literature organization, code drafting, diagnostic review, and prose editing; the human author is responsible for source selection, mappings, protocol choices, statistical specifications, citation verification, and all reported claims.

6 Conclusion

Financial LLM-agent papers often identify profitable strategies, and this study does not dispute returns measured under their stated tasks and benchmarks. Within the 50 retained common-task reconstructions from 40 papers, median annualized residual return falls from 9.97% under CAPM to -1.69% under the market plus JKP132, and the Holm-adjusted discovery count falls to zero. The median reconstruction has 0.81 absolute correlation with its closest JKP factor, with the strongest concentration in low-risk, 52-week-high, volatility, quality, momentum, and valuation families. These findings do not rank the scientific quality, methodological novelty, or practical value of the 98 screened works.

The evidence supports a methodological recommendation rather than a causal verdict about agents. Claims of incremental LLM-generated alpha should report broad factor spanning, factor overlap,

multiplicity correction, costs, and dated reproducible outputs. Independent native-agent replications and outcome-blind mapping studies are still needed to determine how broadly the reconstruction-level result extends to the underlying systems.

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